

Team 1 - COLX 585*

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1 Task Description

The task in this project is to perform a membership inference attack (MIA) on finetuned vision language models. More specifically, we aim to predict whether a given sample was included in the models' finetuning dataset by assigning each sample a membership score. This task is important because it helps evaluate whether a model leaks information about its training data, which is closely related to privacy and memorization risks in machine learning systems. Therefore, the goal of this project is to develop a method that can distinguish member samples from non-member samples as accurately as possible and submit these predictions to the Kaggle competition.

2 Methods

2.1 M⁴I-CLIP: Adapted Multi-modal Membership Inference Attack

Our method adapts the two core attack strategies introduced in the M4I paper by Hu et al. (2022), namely metric-based and feature-based membership inference, to the Kaggle VLM membership inference task. Instead of following the original architecture used in that work, which relied on a ResNet-152 image encoder, an LSTM decoder, and shadow models, we adapt both strategies to a modern vision-language setting by using CLIP as the cross-modal feature extractor and SmolVLM as the target generative model. This adaptation better matches the structure of contemporary vision-language models and allows the attack to be implemented directly on our target system.

For the metric-based branch, we generate text from the fine-tuned SmolVLM using greedy decoding and compare the generated text with the ground-truth answer using two similarity measures: ROUGE-L F1 and token overlap ratio. The token overlap ratio is implemented as a Jaccard-style similarity over sets of lowercased whitespace-tokenized words. We do not use BLEU or BERTScore in the final implementation.

For the feature-based branch, the original M4I framework projects images and texts into a shared embedding space using a custom multimodal feature extractor and then measures the distance between the two modalities. In our adaptation, we replace this handcrafted extractor with OpenAI CLIP ViT-B/32, which provides a strong off-the-shelf shared embedding space for image-text alignment. For each sample, we compute the cosine similarity between the image embedding and the ground-truth text embedding, the image embedding and the generated text embedding, and the ground-truth text embedding and the generated text embedding. We also compute a derived feature, `clip_sim_gap`, defined as the difference between image-generated similarity and image-ground-truth similarity. This replaces Euclidean distance in a learned custom space with cosine similarity in CLIP space.

All extracted features are combined and fed into an XGBoost meta-classifier, consistent with our Milestone 5 pipeline. We evaluate two variants of this attack. The first is the full M4I-CLIP attack, which includes both generation-based and CLIP-based features. This version uses seven features: `rouge_l`, `token_overlap`, `clip_img_text_sim`, `clip_img_gen_sim`, `clip_text_gen_sim`, `clip_sim_gap`, and `gen_length`. The second is a fast CLIP-only attack, which avoids VLM generation and instead combines lightweight CLIP features with precomputed loss-based statistics from Milestone 5. This faster version uses three CLIP-based features—`clip_img_text_sim`, `clip_img_question_sim`, and `clip_text_complexity`—together with up to seven loss-based features: `loss`, `perplexity`, `min_k_prob`, `max_token_loss`, `std_token_loss`, `zlib_ratio`, and `num_tokens`. Here, `clip_text_complexity` is implemented as the norm of the CLIP text embedding and is used as a lightweight proxy feature.

In terms of performance, the full M4I-CLIP attack achieved a Kaggle AUC of 0.80560, while the fast CLIP-only variant achieved a nearly identical Kaggle AUC of 0.80537. On the validation set, the full version achieved a debug AUC of 0.8487

*Repository: <https://github.ubc.ca/snalyf/TFC>

and a TPR@FPR= 0.1 of 0.3333, while the fast version achieved a validation AUC of 0.8093 and a TPR@FPR= 0.1 of 0.3000. Although the full method performs slightly better, the difference is very small, indicating that much of the useful membership signal can already be captured by CLIP-based alignment features without requiring expensive text generation.

2.2 Neighborhood + MIA Combined Attack

Our best-performing method combines two complementary membership inference strategies: a three-layer MIA attack and a dual-model neighborhood attack. The key insight is that these two approaches capture orthogonal signals—the MIA attack measures *absolute memorization* through loss comparisons between the fine-tuned and base models, while the neighborhood attack measures *relative sensitivity* by observing how model confidence changes under input perturbation. Combining both signals into a single linear classifier yields a substantial improvement over either method alone.

2.2.1 Three-Layer MIA Attack

The first component extracts three features by probing the fine-tuned model’s behavior under different conditions. All features are computed using the cross-entropy loss on assistant (caption) tokens only, following SmolVLM’s chat template.

The strongest feature is the **loss ratio**, defined as $\text{loss_ratio} = \text{loss}_{\text{base}} / (\text{loss}_{\text{ft}} + \epsilon)$, where $\text{loss}_{\text{base}}$ and loss_{ft} are the cross-entropy losses from the base and fine-tuned models respectively. The base model serves as a control: it eliminates the confounding effect of inherently easy or hard samples, isolating the differential signal caused by memorization. Members exhibit a lower loss ratio (mean 1.23) compared to non-members (mean 1.57), because fine-tuning disproportionately reduces the loss for memorized pairs. This feature alone achieves a univariate AUC of 0.7929 on the validation set.

The second feature is **caption contrast**, defined as $\text{caption_contrast} = \frac{1}{N} \sum_{i=1}^N \text{loss}(x, c_i^{\text{shuf}}) - \text{loss}(x, c^{\text{orig}})$. For each image, the original caption is replaced with N randomly sampled captions from the dataset pool, and the average loss increase is measured. The intuition is that a memorized (image, caption) binding should produce a large gap between the original and shuffled caption losses. In practice, this feature achieves only AUC = 0.5085 because the VLM generalizes well

to any correct image-caption pairing, making the gap insufficiently specific to memorization.

The third feature is **corruption sensitivity**, defined as $\text{corruption_sensitivity} = -(\text{loss}_{\text{corrupted}} - \text{loss}_{\text{clean}})$, where the image is corrupted with Gaussian blur (radius 2) and additive Gaussian noise ($\sigma = 0.05$). The hypothesis is that a model tuned to specific pixel patterns should be more sensitive to image perturbation for memorized samples. However, this feature achieves only AUC = 0.5203, as SmolVLM’s ViT vision encoder is inherently robust to such perturbations, producing negligible loss changes across all samples (std = 0.015).

2.2.2 Dual-Model Neighborhood Attack

The second component, adapted from our text-only neighborhood attack in Milestone 3, generates perturbed “neighbors” of the input in both text and image modalities and measures how the loss changes under perturbation using both the fine-tuned and base models.

For **text neighborhood** features, we perturb the caption by randomly dropping 10% of words, generating 5 neighbors per sample while keeping the image fixed. We compute four features: `text_nb_gap_ft` (mean perturbed loss minus original loss on the fine-tuned model), `text_nb_gap_base` (the same on the base model), `text_nb_std_ft` (standard deviation of perturbed losses), and `text_nb_relative_gap` (the difference between the fine-tuned and base gaps). The relative gap isolates the fine-tuning-specific sensitivity to text perturbation.

For **image neighborhood** features, we perturb the image using three strategies cycled across 5 neighbors: Gaussian noise ($\sigma = 25.0$), MAE-style patch masking (15% of 32×32 patches), and random crop-resize (70–90% crop ratio). We compute the analogous four features: `img_nb_gap_ft`, `img_nb_gap_base`, `img_nb_std_ft`, and `img_nb_relative_gap`. Among all neighborhood features, `img_nb_std_ft` is the strongest (AUC = 0.5804), as a higher standard deviation across perturbation types indicates the model’s sensitivity varies in a way consistent with overfitting to specific visual features.

We also compute three **cross-modal binding** features: `cross_modal_gap` (text gap minus image gap), `binding_score` (product of both gaps), and `total_nb_gap` (sum of both gaps). The binding score captures whether the model is simultaneously

sensitive to perturbations in both modalities, which is expected only for jointly memorized (image, caption) pairs.

Finally, we include four **base loss** features (loss_ft, loss_base, ref_diff, ref_ratio) and two **text statistics** (text_len, unique_ratio).

2.2.3 Combined Pipeline and Classifier

All 20 features (3 from the MIA attack and 17 from the neighborhood attack) are standardized using StandardScaler and combined into a Logistic Regression classifier with class_weight='balanced' to handle the 1:3 class imbalance (1500 members vs. 4500 non-members in the training set).

We chose Logistic Regression over XGBoost because, despite nearly identical AUC (0.8486 vs. 0.8490), Logistic Regression achieves a higher TPR@FPR= 0.1 (0.3884 vs. 0.3719). With only 20 features and 6000 training samples, the linear model generalizes better and produces better-calibrated probabilities in the low false-positive-rate regime, which is critical for membership inference where high-confidence predictions matter most.

We also evaluated adding CLIP-based features from our M⁴I-CLIP attack (clip_img_text_sim, clip_sim_gap, etc.). While clip_img_text_sim is individually strong (AUC = 0.8046), it is highly collinear with loss_ratio—both capture image-text alignment quality through different mechanisms. Adding CLIP features reduced TPR@FPR= 0.1 from 0.3884 to 0.2727, as the redundant signal diluted the loss ratio’s effectiveness in the high-confidence region. CLIP features were therefore excluded from the final submission.

Forward feature selection using 5-fold cross-validation on the training set identified 14 features as the CV-optimal subset (CV AUC = 0.8119). However, the full 20-feature model achieved higher validation AUC (0.8486 vs. 0.8225 for the selected subset), suggesting that the additional features provide useful regularization at test time despite marginal individual discriminative power.

2.2.4 Results

On the validation set (500 samples; 121 members, 379 non-members), our combined method achieves the best overall performance:

The per-feature univariate AUC analysis confirms that loss_ratio (AUC = 0.7929) is the dom-

Method	AUC	TPR@FPR=0.1
MIA only	0.8154	0.3554
Neighborhood only	0.6305	0.1653
MIA + CLIP	0.8163	0.2727
MIA + Neighborhood + CLIP	0.8322	0.2727
MIA + Neighborhood	0.8486	0.3884
MIA + Neighborhood	0.8490	0.3719

Table 1: Validation set performance across feature combinations and classifiers.

inant signal, followed by img_nb_std_ft (AUC = 0.5804) and text_nb_relative_gap (AUC = 0.5395). Although the neighborhood features are individually weak, their combination with the MIA features improves AUC from 0.8154 to 0.8486 and TPR@FPR= 0.1 from 0.3554 to 0.3884, demonstrating that perturbation-based sensitivity signals provide complementary information to the absolute loss differential captured by the loss ratio.

2.3 Ultimate Hybrid MIA: Multi-Signal Ensemble Attack

Our method combines four orthogonal signal families into a single meta-classifier, building on and supersetting the M⁴I-CLIP approach described in Section 2.1. While M⁴I-CLIP captures membership signals primarily through CLIP cosine similarity (a data-level signal), our method adds model-level signals by comparing the finetuned and base SmolVLM models, inspired by the Likelihood Ratio Attack (LiRA) framework from Carlini et al. (2022) and the generation consistency insight from Hu et al. (2025).

2.3.1 Signal A: Metric-Based Features

Following the M⁴I-CLIP approach, we generate text from the finetuned SmolVLM using greedy decoding and compare it to the ground-truth answer. We extend the original feature set by adding multiple ROUGE variants and BLEU scores: rouge_1, rouge_1, rouge_2, token_overlap, bleu_1, bleu_2, and gen_length. The additional metrics provide complementary signals, as ROUGE captures recall-oriented overlap at different n-gram granularities while BLEU captures precision-oriented overlap.

2.3.2 Signal B: CLIP Embedding Features

Identical to the M⁴I-CLIP implementation, we use OpenAI CLIP ViT-B/32 to compute cosine similarities between image, ground-truth text, and

generated text embeddings. The four features—`clip_img_text_sim`, `clip_img_gen_sim`, `clip_text_gen_sim`, and `clip_sim_gap`—capture data-level alignment quality.

2.3.3 Signal C: LiRA Cross-Model Comparison

The key addition in our method is the Likelihood Ratio Attack, which compares the finetuned model’s token-level statistics against the base (pre-finetuning) SmolVLM on each sample. The intuition is that if a sample was in the finetuning set, the finetuned model’s loss should have dropped more than for a non-member, and this differential is detectable by comparing against the base model’s loss on the same sample.

For each model (finetuned and base), we perform a forward pass and extract six token-level features: mean cross-entropy loss, perplexity, Min-K% Prob (the average log-probability of the bottom 20% hardest tokens, following Shi et al. (2024)), maximum single-token loss, standard deviation of token losses, and zlib-normalized loss ratio. From these paired measurements, we derive eight cross-model features:

- $\text{loss_ratio} = \text{loss}_{\text{base}} / (\text{loss}_{\text{ft}} + \epsilon)$ — higher values indicate greater loss reduction from finetuning, suggesting membership.
- $\text{loss_diff} = \text{loss}_{\text{base}} - \text{loss}_{\text{ft}}$ — the absolute loss improvement.
- $\text{min_k_diff} = \text{min_k_ft} - \text{min_k_base}$ — whether even the hardest tokens improved after finetuning.
- $\text{perplexity_ratio} = \text{ppl}_{\text{base}} / \text{ppl}_{\text{ft}}$ — perplexity-space comparison.
- max_loss_diff , std_loss_diff , zlib_ratio_diff — analogous differentials for other token-level statistics.
- $\text{norm_loss_improvement} = (\text{loss}_{\text{base}} - \text{loss}_{\text{ft}}) / \text{loss}_{\text{base}}$ — relative loss drop.

This signal family is conceptually similar to the loss ratio used in the Neighborhood + MIA method (Section 2.2), but extracts a richer set of cross-model features beyond the single loss ratio, including Min-K% differentials and token-level variance changes.

2.3.4 Signal D: Generation Consistency

Inspired by the set-level consistency observation from Hu et al. (2025), who found that members exhibit greater sensitivity to temperature changes, we measure output consistency by generating multiple responses with sampling (temperature = 1.0) and computing pairwise ROUGE-L similarity across outputs. Members are expected to produce more consistent descriptions because the model memorized a specific response for those image-text pairs. We extract two features: `gen_consistency_mean` (average pairwise similarity) and `gen_consistency_std` (variance in pairwise similarity).

2.3.5 Meta-Classifer

All features (up to 33) are standardized with `StandardScaler` and fed into an `XGBoost` classifier with 500 estimators, max depth of 4, learning rate of 0.05, and column subsampling ratio of 0.7. The script includes a built-in signal family ablation that reports AUC for each family in isolation and combined, using Logistic Regression for fair comparison.

2.3.6 Results

Method	Val AUC	TPR@FPR=0.1
M ⁴ I-CLIP	0.8093	0.3000
MIA + Neighborhood	0.8486	0.3884
MIA-LiRA	0.8061	0.2633

Table 2: Validation set performance comparison across methods.

3 Literature Review

Paper 1: Practical Membership Inference Attacks against Fine-tuned Large Language Models via Self-prompt Calibration

This research investigates the privacy risks associated with Large Language Models that have been fine-tuned on private data while maintaining high generalization capabilities. The main task addressed by the authors is membership inference, where an attacker seeks to determine whether a specific data record was part of a model training set. Unlike traditional attacks that rely on visible overfitting, this work focuses on models that appear overfitting-free. To evaluate their methodology, the researchers utilize three prominent benchmarks: the **AG News** classification dataset, the

XSum summarization dataset, and the **SQuAD** question-answering dataset. The authors propose a novel framework called **SPV-MIA** (Self-calibrated Probabilistic Variation). This model leverages a self-prompting strategy where the target model is queried to generate its own training-like data to build a reference model for calibration. This allows for the calculation of probabilistic variation, a metric that serves as a more reliable signal for memorization than raw loss. The main contribution of this paper is the demonstration that memorization is an inevitable byproduct of the training process that can be exploited even without access to the original training distribution. By automating the creation of reference datasets via self-prompting, the authors show that practical Large Language Models remain vulnerable to sophisticated privacy attacks.

Citation: Fu, W., Wang, H., Gao, C., Liu, G., Li, Y., & Jiang, T. (2024). Practical Membership Inference Attacks against Fine-tuned Large Language Models via Self-prompt Calibration. Advances in Neural Information Processing Systems, 37 (NeurIPS 2024). arXiv:2311.06062.

Paper 2: Min-K%++: Improved Baseline for Detecting Pre-training Data from Large Language Models

This research addresses the detection of pre-training data within Large Language Models to investigate critical issues such as copyright infringement and test set contamination. The main task involves determining whether a specific data record was utilized during the initial pre-training phase of a model, a problem that is complicated by the massive scale of modern training corpora. The researchers evaluate their methodology using the **WikiMIA** and **MIMIR** benchmarks across multiple model families, including **LLaMA**, **Pythia**, and the state-space architecture **Mamba**. The authors propose a novel, reference-free method called **Min-K%++**, which builds on the insight that training samples tend to form local maxima in the likelihood space. By normalizing token probabilities with statistics derived from the entire categorical distribution of the vocabulary, the method identifies training data more reliably than existing heuristics. The main contribution of this work is the development of a theoretically grounded detection tool that achieves state-of-the-art performance without the computational burden of a reference model, providing a robust baseline for evaluating data trans-

parency and privacy in large-scale language modeling.

Citation: Zhang, J., Sun, J., Yeats, E., Ouyang, Y., Kuo, M., Zhang, J., Yang, H. F., & Li, H. (2025). Min-K%++: Improved Baseline for Pre-Training Data Detection from Large Language Models. In The Thirteenth International Conference on Learning Representations (ICLR 2025).

4 Contributions

Member	Contributions	Percentage
Tianhao Cao	Method	25%
Yusen Huang	Method	25%
Marco Wang	Method	25%
Darwin Zhang	Literature Review	25%

Table 3: Team contributions for Milestone 6.